Mohamed El Khoudimi

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Technical Skills

Languages: Python, C++20/17, C, SQL, Bash; data structures, algorithms and object-oriented design
Shipping and operations: AWS CDK, Lambda, CloudWatch alarms, SNS, KMS, IAM; OpenShift (Kubernetes) readiness and liveness probes, HashiCorp Vault; Docker, Linux, GitHub Actions CI
Backend and data: FastAPI, REST APIs, Pydantic, Redis event bus, TimescaleDB (PostgreSQL), SQL, pytest, k6 load tests
AI-assisted development: GitHub Copilot, Cursor, Claude Code, ChatGPT; Amazon Bedrock, Databricks AI, Atlassian Rovo
Spoken: English, French, Arabic (fluent)

Experience

BMO Financial Group Toronto, ON
AI & Software Engineer *Sep 2026 - Dec 2026*

- Shipped automated database health checks to production on AWS (CDK, Lambda, CloudWatch, RDS APIs, SNS, KMS, IAM), with a multi-instance report and a change-managed release from a verified commit.
- Containerized a FastAPI agent API and deployed it on OpenShift (Kubernetes) with readiness and liveness probes and Vault-managed secrets.
- Built an AI incident-triage service (Python, FastAPI, RAG) used by ~30 engineers; cut median initial diagnosis from 28 to 15 min (46%) across 240 pilot incidents, with read-only access and 100% human approval for operational actions.
- Analyzed ~1,500 historical incidents for recurring failure patterns, informing fixes that cut one targeted incident category by ~15%; automated or accelerated ~100 operational events a month (~900 hours a year saved, estimate).

NTT Corporation Moncton, NB
Fraud Analyst & Financial Reporting Analyst *May 2022 - Sep 2024*

- Investigated 200+ financial and fraud cases under SLA and compliance rules; automated Python reporting, cutting manual reporting effort by ~60%, and improved handover efficiency by 35%.

Volunteer Tutor May 2025 - Present

- Tutored ~10 students one-on-one (all referrals) in algorithms, operating systems, distributed systems and C++; every student finished at 80%+, most above 90%.

Projects

Aster Market Engine / *C++20, CMake, Python* github.com/mtk1606/aster-market-engine

- Built a C++20 Nasdaq TotalView-ITCH 5.0 replay engine with a full-depth price-time FIFO order book (solo, public repo; 10/10 tests, 5/5 CI jobs) and benchmarked the engine on a real session (Jan 30 2019: 368,366,634 messages, 11.25 GB).
- Split cost by stage and I/O path (median of 3 runs): decode only 7.547M msg/s with read() vs 7.255M with mmap; full replay with 363,118,215 book updates 1.341M msg/s with read() (275 s, ~243 MiB max RSS) vs 1.354M with mmap.
- Correlated the C++ book against an independent Python reference model: matched over the first 20M messages at 10 checkpoints, then AAPL at frame 120M (39,826 orders) and MSFT at frame 300M (19,502 orders), zero diffs.

AACP Event-Driven Fraud Engine / *Python, FastAPI, Redis, k6* github.com/mtk1606/aacp-fraud-engine

- Benchmarked ~395 req/s sustained under k6 on a real-time FastAPI and LangGraph control plane streaming via a Redis event bus into TimescaleDB.
- Implemented a PID feedback loop that tunes anomaly thresholds toward a 2% false-positive-rate target; ingestion fails open and high-risk events fail safe to human review.

QSE Trading Engine (co-built with Marcus Myrick) / *C++17/20, Linux*

- Led cross-platform determinism testing for a co-built C++17/20 event-driven trading engine on Linux.

Education

Université de Moncton 2024
Bachelor of Engineering (B.Eng.), Mechanical Engineering, GPA 3.4/4.0

Georgian College Expected Apr 2027
Diploma in Computer Programming, GPA 3.7/4.0